

Here is a concise English summary of the provided Persian thesis, written for a Master's level Computer Science audience (e.g., for the University of Bonn).

Title: Smartphone-Based Human Activity Recognition Using Deep Learning Methods

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Abstract & Motivation

This thesis investigates the problem of Human Activity Recognition (HAR) using data collected from smartphone sensors (accelerometer and gyroscope). The work leverages the UCI HAR dataset, which captures six activities: walking, walking upstairs, walking downstairs, sitting, standing, and laying. While traditional machine learning relies on hand-crafted feature extraction, this research explores various Deep Learning (DL) architectures capable of automatic feature learning from raw time-series data.

Methodology

The study implements and evaluates several deep learning models:

1. **1D Convolutional Neural Networks (CNN):** Used for local feature extraction from sensor signals.
2. **Long Short-Term Memory (LSTM):** A recurrent neural network designed to capture temporal dependencies in sequential data.
3. **Bidirectional LSTM (Bi-LSTM):** Processes data in both forward and backward directions to utilize past and future context.
4. **Stacked LSTM:** Multiple LSTM layers to increase model depth.
5. **Hybrid CNN-LSTM:** Combines convolutional layers for spatial feature extraction with LSTM layers for temporal sequence modeling.

Data Preprocessing

Raw signals were preprocessed using:

- **Filters:** Median filter and Butterworth low-pass filter to remove noise.
- **Normalization:** Z-score normalization (mean and standard deviation).
- **Windowing:** Data segmented into fixed-length windows (128 samples / 2.56 seconds).

Key Results

The models were evaluated using Accuracy, Precision, Recall, and F1-score via K-Fold cross-validation.

Model Type	Accuracy	Precision	Recall	F1-Score
CNN	93.31%	93.32%	93.45%	93.29%
LSTM	94.45%	94.50%	94.62%	94.50%
Bidirectional LSTM (Stacked)	94.95%	95.10%	95.13%	95.09%
CNN-LSTM (Deep)	94.38%	94.44%	94.54%	94.49%

Conclusion

The **Bidirectional Stacked LSTM** achieved the highest overall performance, with an accuracy of 94.95% and an F1-score of 95.09%. The study demonstrates that bidirectional architectures, which leverage both past and future signal context, significantly outperform standard LSTMs and CNNs for HAR tasks.

Future Work

- Expand the activity set beyond the six standard categories.
- Propose novel hybrid architectures (e.g., Graph Neural Networks) to model relationships between sensor nodes.
- Further improve noise reduction techniques for sensor data.

link to the full thesis:

https://drive.google.com/file/d/1um3GI7WPvLyYBIP1THEbrmdhf9eqg_ZC/view?usp=sharing